Graphical Abstract

Implementation and Validation of 1D Return Mapping Algorithms for Plasticity and Viscoplasticity

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Highlights

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- Research highlight 2

Implementation and Validation of 1D Return Mapping Algorithms for Plasticity and Viscoplasticity

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Abstract

Abstract text.

Keywords: Return mapping, Plasticity, Viscoplasticity, Perzyna model, Backward Euler

1. Introduction

- Motivation (why computational plasticity matters)
- Return mapping as standard approach in FEM
- Ibjectives of month 1 work
- Organization of this report

2. Theoritical background

2.1. Rate-Independent Plasticity

- Yield function: $f(\sigma) = |\sigma| - \sigma_y$
- Flow rule with normality
- Kuhn-Tucker conditions
- Key equation: $\Delta\lambda = f^{trial}/E$

2.2. Viscoplasticity

- Perzyna mode: $\dot{\varepsilon}^{vp} = \frac{1}{n}\left\langle \frac{f}{\sigma_y} \right\rangle^n sign(\sigma)$
- Physical meaning of η and n
- Limiting case $n \to 0$ (recovers rate indenpendence)

3. Implementation

3.1. Algorithm Structure

- Elastic predictor
- Yield check
- Plastic corrector
- Figure
  - Simple flowchart to illustrate the process of return mapping

3.2. Code Organization

Algorithm 1: MAX finds the maximum number

Input: Current step material data $(\sigma_n, \varepsilon_n, \varepsilon_n^p)$
Output: Updated material data $(\sigma_{n+1}, \varepsilon_{n+1}, \varepsilon_{n+1}^p)$

```
max ← a_1
for i ← 2 to n do
    if a_i > max then
        max ← a_i
    end
end
return max
```